No Overlap, No Invariance: An Identifiability Boundary for Anchor-Free Disentanglement, and the Functionals That Survive It

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Abstract

Many applications ask for a representation that is invariant to which domain a sample came from while retaining the biological or semantic factor of interest, and ask for it *without* anchors: no paired samples, no shared labels, only two unaligned corpora. We study when this is achievable at all. We give an overlap-based boundary condition, that anchor-free invariance is identified only where the two domains share support, and we test it in a real regime of extreme scale and near-disjoint support: a 2,615,898-cell primary reference and a 1,767,346-cell in-vitro query. A shared-latent contrastive model fails decisively and reproducibly: a held-out adversary recovers the domain from the putatively invariant latent at ≥ 0.998 balanced accuracy, unchanged when the salient capacity is tripled and when the decoder is conditioned on domain, while training converges normally. The failure is structural rather than optimization-related, because the encoder observes the domain directly and decoder-side conditioning cannot remove what the encoder has already written. We then ask what remains estimable when the per-sample invariant coordinate is not identified, and show that translation-invariant functionals are the natural answer: pairwise sliced-Wasserstein distances are invariant to an offset shared by the compared populations, which lets relative geometry be recovered without ever constructing an invariant coordinate. This route has its own precondition, that the offset be constant across the compared populations, and we test it directly rather than assuming it. It fails: per-population offsets are mutually inconsistent (mean cosine 0.15, against 0.855 within domain) and comparable in magnitude to the between-population separation, so individual distances are confounded. What survives is weaker but real: the induced *ordering* still recovers the reference geometry (distance-matrix $r = 0.90$). The result is a graded answer, coordinates are not identified, distances are biased, orderings are estimable, and a reusable recipe for deciding which of the three a given problem admits.

1 Introduction

A recurring pattern across applied machine learning is the desire for a representation that discards *where the data came from* while keeping *what the data is.* In domain adaptation this is an invariant feature map; in generative modeling it is a disentangled latent; in scientific data analysis it is the hope of subtracting a measurement or environment effect to leave the underlying biology. The version of this problem we study is the hardest common one, which we call *anchor-free*: we are given two unaligned corpora from two domains, with no paired samples, no shared instance labels, and no overlap-by-construction, and we are asked to learn a latent coordinate that is invariant to domain while remaining informative about the factor of interest.

Practitioners routinely attempt this with shared-latent generative models, in which a background latent is shared across domains and a salient latent absorbs what is unique to one of them, optionally with the decoder conditioned on a domain covariate. The implicit assumption is that the domain difference is a *separable nuisance subspace*, something a model can be induced to route away from the shared coordinate. We ask when that assumption can hold, and what to do when it does not.

Our answer has three parts, and the third is the one we find most useful.

An overlap boundary. We state, and then test, the condition under which anchor-free invariance is identified at all. If the two domains occupy essentially disjoint regions of input space, then domain is almost definitionally decodable from any representation that supports reconstruction, and, more importantly, there is no region in which a *common* coordinate is even constrained: nothing ties the axis learned on one domain to the axis learned on the other. Anchor-free invariance is thus identified only on the overlap, and the failure is not a matter of capacity or regularization.

A decisive empirical instance. We test this in a regime that is unusually favorable for detecting the failure: two corpora totaling 4.4 million samples with rich, well-characterized structure, where the domains are known a priori to be near-disjoint. A shared-latent contrastive model yields an adversary balanced accuracy of 0.998 on the supposedly invariant latent. Tripling the salient capacity gives 0.999; conditioning the decoder on the domain, the textbook remedy for a nuisance factor confounded with a condition, gives 0.998. Training converges in every case, so this is not an optimization failure. We are explicit about what this does and does not establish: the adversary shows the domain difference is *decodable*, not what it is *made of*.

What remains estimable, and a test of it. The more useful question is what survives non-identifiability. We observe that the target of many downstream analyses is not the per-sample coordinate but a *relative* quantity, and that relative quantities can be invariant to the very offset that defeats the coordinate: pairwise sliced-Wasserstein distances satisfy $W(\mathcal{A}+c, \mathcal{B}+c) = W(\mathcal{A}, \mathcal{B})$, so a domain offset shared by both compared populations cancels exactly. This yields estimable functionals without any invariant coordinate. But the cancellation has a precondition, that the offset really is shared, and we treat that as an empirical claim to be tested rather than assumed. Testing it, we find it false in our setting: per-population offsets are mutually inconsistent and as large as the between-population separation. The distances are therefore biased. Yet the induced ordering is not: the recovered ranking reproduces the reference geometry at $r = 0.90$. The practical output is a three-level ladder, coordinates, distances, orderings, and a procedure for determining which level a given problem supports.

Contributions.

- An overlap-based boundary condition for anchor-free invariance, with the mechanism made explicit (encoder-visible domain; decoder conditioning is insufficient by construction).
- A large-scale, reproducible negative result establishing the boundary empirically, robust across capacity and decoder-conditioning interventions, with the interpretive scope carefully separated from the measurement.
- A positive counterpart: translation-invariant functionals as the estimable object under non-identifiability, together with a direct test of their precondition, yielding the graded conclusion that orderings survive where distances do not.
- A diagnostic for a confound that afflicts any such study, distinguishing *reference incompleteness* from *genuine divergence* when a query is declared out-of-distribution, including a calibration trick that remains valid when the naive control fails.

2 Setup and problem statement

Anchor-free invariance. Let P and Q be distributions over an input space $\mathcal{X}$, corresponding to two domains (in our instance, primary tissue and an in-vitro system). We observe two unaligned samples, $X_P \sim P$ and $X_Q \sim Q$, with no correspondence between them. Each sample carries an unobserved factor of interest y (here, cell lineage) whose meaning is intended to be shared across domains. We seek an encoder $q(z \mid x)$ such that z is (i) uninformative about the domain indicator $d \in \{P, Q\}$, and (ii) informative about y, using no paired samples and no shared labels.

The shared-latent contrastive family. A widely used approach factorizes the latent into a shared (background) part z_{sh} and a salient part z_{sal} that is active only for the target domain, so that domain-specific variation is absorbed by z_{sal} and z_{sh} carries the common factor. A decoder $p(x \mid z_{\mathrm{sh}}, z_{\mathrm{sal}}, d)$ may additionally be conditioned on d. We instantiate this family with contrastiveVI (Weinberger et al., 2023), built on the variational single-cell framework of Lopez et al. (2018); the argument below is about the family, not the implementation.

Measurement. We quantify invariance by the *decodability* of the domain from the shared latent: a cross-validated classifier $\hat{d}(z_{\mathrm{sh}})$, scored by balanced accuracy. A value of 0.5 indicates invariance; a value near 1 indicates the domain is fully recoverable. This is used strictly as an evaluation probe, never as a training signal.

An overlap boundary condition (informal). Consider the support overlap $\mathcal{O} = \mathrm{supp}(P) \cap \mathrm{supp}(Q)$. Two observations follow, and together they delimit what anchor-free invariance can mean.

(a) Off the overlap, invariance is unconstrained. If a region of input space is visited by only one domain, then no term in an anchor-free objective relates the coordinate assigned there to the coordinate assigned to any Q-region: the two domains' latent axes may be arbitrarily rotated, scaled, or translated relative to one another without penalty. A "common" axis is therefore not identified; it is chosen by inductive bias, not by data. This is the anchor-free analogue of the familiar point that unsupervised disentanglement is unidentifiable without inductive bias or supervision (Locatello et al., 2019), and of the classical result that nonlinear ICA is unidentifiable in general (Hyvärinen & Pajunen, 1999), with the additional twist that here even a shared parameterization does not help, because the shared parameters are never jointly constrained.

(b) When $\mathcal{O}$ has negligible mass, domain is decodable by construction. If P and Q are near-disjoint, then x determines d almost surely, so any z that retains enough information to reconstruct x retains d. Crucially, this is a property of the *encoder's input*, and conditioning the *decoder* on d cannot repair it: the decoder covariate can explain away the domain-specific *mean shift* in the likelihood, but it does not remove the domain information the encoder has already written into z. This asymmetry is why the standard batch-conditioning remedy is expected to fail here.

We stress that (b) alone is nearly a tautology, and we do not present it as a surprise. The substantive claim is (a) plus (b) together: anchor-free invariance is identified only on the overlap, so the interesting question for a practitioner is not "did my model fail" but "does my problem have enough overlap to admit the target at all, and if not, what should I estimate instead." Sections 4 through 6 answer that in a concrete, high-stakes setting.

3 A near-disjoint benchmark at atlas scale

We need a testbed where (i) the two domains are large, (ii) their supports are known to be near-disjoint, and (iii) the factor of interest has independently known structure so that success and failure are distinguishable. Single-cell transcriptomics of in-vitro organoids versus primary fetal tissue provides exactly this (Fig. 1).

Domains. The primary domain is a harmonized reference of 2,615,898 cells $\times$ 61,497 genes assembled from a first-trimester human brain atlas (Braun et al., 2023) and a fetal-body atlas (Cao et al., 2020), retrieved with raw counts from a public single-cell platform (CZI Cell Science Program, 2025). The query domain is an integrated in-vitro atlas of 1,767,346 organoid cells spanning 27 differentiation protocols (He et al., 2024). The factor of interest, y, is cell lineage, for which curated ontology labels exist in both domains.

Why near-disjoint. The two domains differ simultaneously in biology (an in-vitro stress program absent from tissue) and in measurement (laboratory, dissociation, platform, ambient RNA). We verify the consequence rather than assume it: Section 7 shows that even query cells that map *confidently* to this reference remain far from an *independent* reference of the same tissue, i.e. the domain offset displaces essentially all query cells.

Mapping and per-sample scores. We map the query onto the reference by variational reference surgery (Lopez et al., 2018; Xu et al., 2021; Lotfollahi et al., 2022; Gayoso et al., 2022), obtaining a reference model (evidence lower bound 695.6), a label-aware refinement (736.1), and a query projection (920.2). Donor-level integration improves cross-batch mixing while preserving structure (integration LISI $6.10 \to 8.10$; label LISI $1.31 \to 1.44$) (Korsunsky et al., 2019; Luecken et al., 2022). Each query sample receives two calibrated scores, an off-manifold distance and a label-assignment entropy, which together partition the query into interpretable strata (Fig. 2): confidently mapped (26.0%), weakly differentiated (15.9%), ambiguous (9.5%), *off-manifold* (38.4%), *off-target* lineages that map confidently to a non-target identity (5.3%), and quality-filtered (4.9%). Full construction details are in Appendix A.

The two strata that matter below are the *off-manifold* stratum, the large ambiguous mass whose interpretation Section 7 addresses, and the *off-target* stratum, a set of well-defined non-target lineages that provides the multi-population geometry used in Sections 5–6.

4 Result 1: anchor-free invariance is not identified in this regime

We train the shared-latent contrastive model with primary cells as background and query cells as target (100,000 each, 2,000 highly variable genes, raw counts), then probe z_{sh} for domain decodability (Fig. 3a).

The result is decisive and intervention-robust. The baseline model ($\dim z_{\text{sal}} = 10$) gives adversary balanced accuracy 0.998. Tripling the salient capacity ($\dim z_{\text{sal}} = 30$), which by the separable-nuisance intuition should give the domain axis somewhere to go, gives 0.999. Registering the domain as a batch covariate on the decoder, the standard remedy, gives 0.998. Training converges normally in all three runs (Fig. 3b), so the failure is not optimization. This is the pattern predicted by Section 2: the encoder sees the domain, and decoder conditioning cannot unsee it (Fig. 3c).

What the probe does and does not establish. We are deliberate here, because the two claims are easy to conflate. The adversary establishes that the domain difference is *decodable* from z_{sh}. It does *not* establish what that difference consists of: our two domains differ in biological state *and* in laboratory, dissociation, platform, and ambient-RNA profile, and any one of these is pervasive enough on its own to make the domain decodable. The defensible claim is therefore methodological, that shared-latent contrastive methods do not yield a domain-invariant coordinate in this regime, whereas attributing the dominant axis to a specific biological cause requires the separate evidence of Section 7. A second scope limit: the donor-level integration used for mapping does not correct the far deeper cross-domain divide, so that divide is exactly what z_{sh} inherits.

Scope of the negative. We tested one model family, thoroughly, under the two interventions its own logic prescribes. We do not claim that *no* method can achieve invariance here; we claim that the shared-latent contrastive family cannot, and we give the structural reason, which is a property of the data (support disjointness) rather than of the estimator. Methods that intervene on the *encoder*, adversarially or through an invariance penalty (Ganin et al., 2016; Arjovsky et al., 2019), are the natural candidates to test next, though the theory of invariant representations under label shift (Zhao et al., 2019) suggests they trade invariance against informativeness rather than escaping the bound. A controlled overlap sweep, in which support overlap is varied and the transition located, is the experiment we would most like to add and is discussed in Section 9.

5 Result 2: translation-invariant functionals remain estimable

Non-identifiability of the coordinate does not imply that nothing is estimable. Many downstream questions are *relative*: not "where is this sample on the invariant axis" but "are these two populations different, and which pairs are more different." Relative quantities can be immune to the offset that defeats the coordinate.

Proposition (translation invariance). For populations $\mathcal{A}, \mathcal{B}$ and a shared offset c, the sliced-Wasserstein distance satisfies

$$W(\mathcal{A} + c,\ \mathcal{B} + c) \ = \ W(\mathcal{A}, \mathcal{B}),$$

since every one-dimensional projection shifts both empirical measures identically (Bonneel et al., 2015; Peyré & Cuturi, 2019). Hence if the domain offset in z_{sh} is a *constant* vector across the compared populations, pairwise distances between them are unaffected by it, even though each population's absolute position is not identified.

This is what makes the analysis possible at all: all compared populations lie in the *same* domain (they are all query samples), so they plausibly share the domain offset, which then cancels.

Instantiation. We compare the off-target lineages of Section 3 in z_{sh}, controlling sample size by a rarefaction-derived minimum ($N_{\min} = 2000$) and calibrating against a matched-N self-distance noise floor; a pair is called only when its bootstrap cross-distance interval clears the floor. Of 17 off-target populations, four are well powered (adrenal-cortical 49,338; skeletal muscle 30,629; fibroblast 6,349; cardiac muscle 2,534). All six pairwise comparisons are real differences, clearing the floor by 14 to 105× (W from 0.32 to 0.99 against floors of 0.007 to 0.026); the remaining 130 small-N comparisons are correctly returned as underpowered (Fig. 4). The recovered ordering is chemically sensible, with the two myogenic populations closest, though we emphasize in Section 6 that separating grossly distinct lineages is a low bar and is an instrument check rather than a finding.

6 Result 3: testing the precondition, and a graded answer

The cancellation in Section 5 requires the offset c to be *the same* for the populations being compared. That is an assumption about data, not a theorem, and it is the assumption on which the entire analysis rests. It is also one that domain knowledge actively calls into question: if part of the domain shift is a stress response that scales with a population's secretory demand, then c should differ by population. We therefore measured it.

Procedure. For each of the four well-powered lineages we estimated the domain offset directly as the difference between the query and reference centroids of that lineage, in a shared log-normalized highly-variable-gene space, using the matched ontology label in both domains (2,500 cells per lineage per domain). We then asked (i) whether these offsets are mutually consistent, (ii) whether they distort the between-population geometry, and (iii) whether the induced ordering survives.

The offsets are not constant. Mean pairwise cosine between per-lineage offsets is 0.15, against 0.855 for a within-domain consistency baseline computed on target-matched types. Only the two myogenic lineages align appreciably (0.55); the fibroblast offset is essentially orthogonal to the others. Moreover, each pairwise offset *difference* is comparable in magnitude to the between-lineage separation itself (ratio ≈ 1). The precondition fails, and not marginally.

Consequence: distances are biased, ordering is not. Because $c_{\mathcal{A}} \neq c_{\mathcal{B}}$, the measured distance carries a $\|c_{\mathcal{A}} - c_{\mathcal{B}}\|$ contribution, and the direction of the between-population difference is substantially rotated: the cosine between the query and reference inter-lineage vectors is only 0.29. Individual distances therefore cannot be read as clean between-population metrics. What does survive is the rank structure: across the six pairs, the query and reference pairwise-distance matrices correlate at $r = 0.90$. The ordering is recovered even though the metric is not.

A three-level ladder. We suggest reporting results at whichever of the following a problem supports, and testing rather than assuming the transition:

1. **Coordinates** require overlap (Section 2); unavailable here.
2. **Distances** require a population-constant offset; testable by the procedure above; unavailable here.

3. **Orderings** require only that the offset perturbation not reverse rank; available here ($r = 0.90$), and sufficient for many practical questions.

The contribution is less the particular verdict than the fact that level 2 is checkable with a short, cheap measurement that practitioners currently omit.

7 Result 4: separating reference incompleteness from genuine divergence

Any study of this kind faces a confound that is rarely tested. When a query sample is declared out-of-distribution with respect to a reference, there are two explanations: the sample is genuinely divergent, or the reference is simply *incomplete* and lacks the corresponding state. Conflating them inflates apparent divergence and, in applications, mislabels perfectly ordinary samples as anomalous. Our benchmark's large off-manifold stratum (38.4%) is exactly such a case.

Structure, not noise. Four diagnostics show the stratum is reproducible and structured rather than an artifact (Fig. 5): off-manifold samples show reduced canonical progenitor identity and elevated stress programs; their frequency rises monotonically with a per-sample complexity proxy ($0.13 \rightarrow 0.78$ across quintiles); it varies nearly 240-fold across the 27 protocols (0.4% to 96.8%); and it concentrates in coherent clusters rather than scattering. The protocol gradient is itself diagnostic: protocols targeting states the reference covers score low, while protocols targeting states it lacks score high, which is the signature of reference incompleteness.

A direct test, and the confound it exposes. The decisive test is whether off-manifold samples map onto an *independent* reference of the same tissue. We projected them, together with a confidently-mapped control, onto an independent atlas (Velmeshev et al., 2023). The naive version of this test fails, and instructively: the domain offset displaces *every* query sample from the independent reference, including the control (median distance 4.0 versus 2.2 within reference), under both projection and joint integration. Absolute "mapped-on" fractions are therefore uninterpretable, and we do not report them. This is the same pervasiveness established in Section 4, now observed from a second direction.

The calibrated contrast. The control, however, *quantifies* the shared displacement, and the same cancellation logic applies: comparing off-manifold samples against the control removes it. Off-manifold samples are significantly further from the independent reference than controls (AUC 0.65, Cliff's $\delta = 0.30$; 70% exceed the control median and 15% lie beyond its 95th percentile; Fig. 6). This is confound-controlled evidence for a genuine divergence component over and above reference incompleteness, and it is obtained precisely by refusing to trust the uncalibrated measurement. We regard the pattern, naive control fails, calibrated contrast survives, as the transferable lesson.

8 Related work

Identifiability of latent variables. That unsupervised disentanglement is unidentifiable without inductive bias or supervision is established both classically, through the non-uniqueness of nonlinear ICA (Hyvärinen & Pajunen, 1999), and empirically at scale (Locatello et al., 2019); identifiability can be restored with auxiliary variables (Khemakhem et al., 2020). Our setting differs in that the sought factor is not a generic latent but the domain indicator, and the obstruction is not the usual rotational ambiguity but the absence of a region in which the two domains' axes are jointly constrained.

Domain adaptation and invariant representations. Classical bounds relate target error to source error plus a divergence between domains (Ben-David et al., 2010a), and are vacuous when that divergence is maximal, which is the regime we study; correspondingly, adaptation is impossible without a distributional-proximity assumption (Ben-David et al., 2010b). Adversarial feature alignment (Ganin et al., 2016) and invariance penalties (Arjovsky et al., 2019) intervene on the encoder rather than the decoder, which our analysis identifies as the necessary locus; nonetheless Zhao et al. (2019) show that enforcing invariance can provably hurt when label distributions differ, so encoder-side methods are not a free escape. The overlap condition we emphasize is the representation analogue of the positivity/overlap assumption in causal inference, which is known to fail silently in high dimensions (D'Amour et al., 2021).

Cross-domain integration in single-cell data. Reference mapping and batch integration are mature (Lopez et al., 2018; Xu et al., 2021; Lotfollahi et al., 2022; Korsunsky et al., 2019; Luecken et al., 2022), and contrastive latent-variable models are the standard tool for isolating condition-specific variation (Weinberger et al., 2023). Our results delimit where that toolchain's invariance claim holds, and offer the translation-invariance route where it does not.

9 Discussion and limitations

What we would add next. The single most valuable addition is a *controlled overlap sweep*: constructing paired corpora whose support overlap can be dialed from complete to disjoint, and locating the transition at which anchor-free invariance stops being recoverable. Our evidence establishes the failure at one extreme point, with the mechanism, but a sweep would convert a boundary *claim* into a measured boundary *curve*. A complementary experiment probes what z_{sh} retains *besides* domain, since a representation from which both domain and lineage are decodable is entangled rather than merely broken, and the two have different remedies.

Limitations. (i) The negative is established for one model family; we argue from a property of the data, but a broader sweep of estimators would strengthen the generalization. (ii) The offset-constancy test is conducted in expression space rather than in z_{sh}, as a tractable proxy; a latent-space version requires reference samples to be passed through the trained model. (iii) Our factor-of-interest labels are themselves model-assigned for the query domain, so the ordering validation inherits any systematic labeling error. (iv) The well-powered populations are grossly distinct lineages, a low bar for the instrument; the ordering result should not be read as a discovery about the underlying biology.

Takeaway. The practical message is not that a particular model failed. It is that anchor-free invariance carries an overlap precondition that is rarely checked, that the estimable object under non-identifiability is a translation-invariant functional rather than a coordinate, and that this functional carries a precondition of its own, which is cheap to test and, in at least one realistic setting, false. Reporting the highest rung of the coordinate/distance/ordering ladder that a problem actually supports seems to us a better default than reporting coordinates and hoping.

Broader impact statement

This work analyzes representation learning on public, de-identified cell-atlas data and introduces no new data collection. Its main societal relevance is indirect and, we believe, positive: it argues against over-reading model-derived "invariant" coordinates in scientific applications, a failure mode that can produce confident but unsupported biological conclusions. We identify no significant risk of harm.

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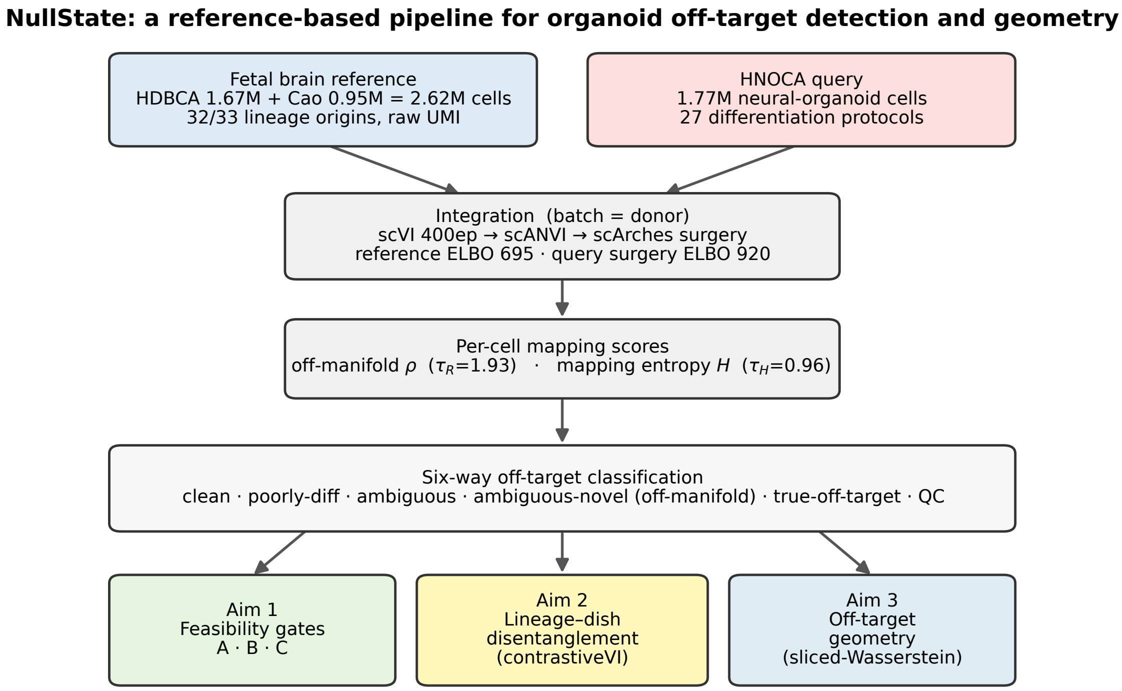

Figure 1: **The benchmark.** Two unaligned corpora, a 2,615,898-cell primary reference and a 1,767,346-cell in-vitro query, are co-embedded by variational reference surgery. Each query sample receives two calibrated scores (an off-manifold distance and a label-assignment entropy) that partition it into interpretable strata. The domains are near-disjoint by construction, which is what makes this a decisive test of the overlap boundary of Section 2.

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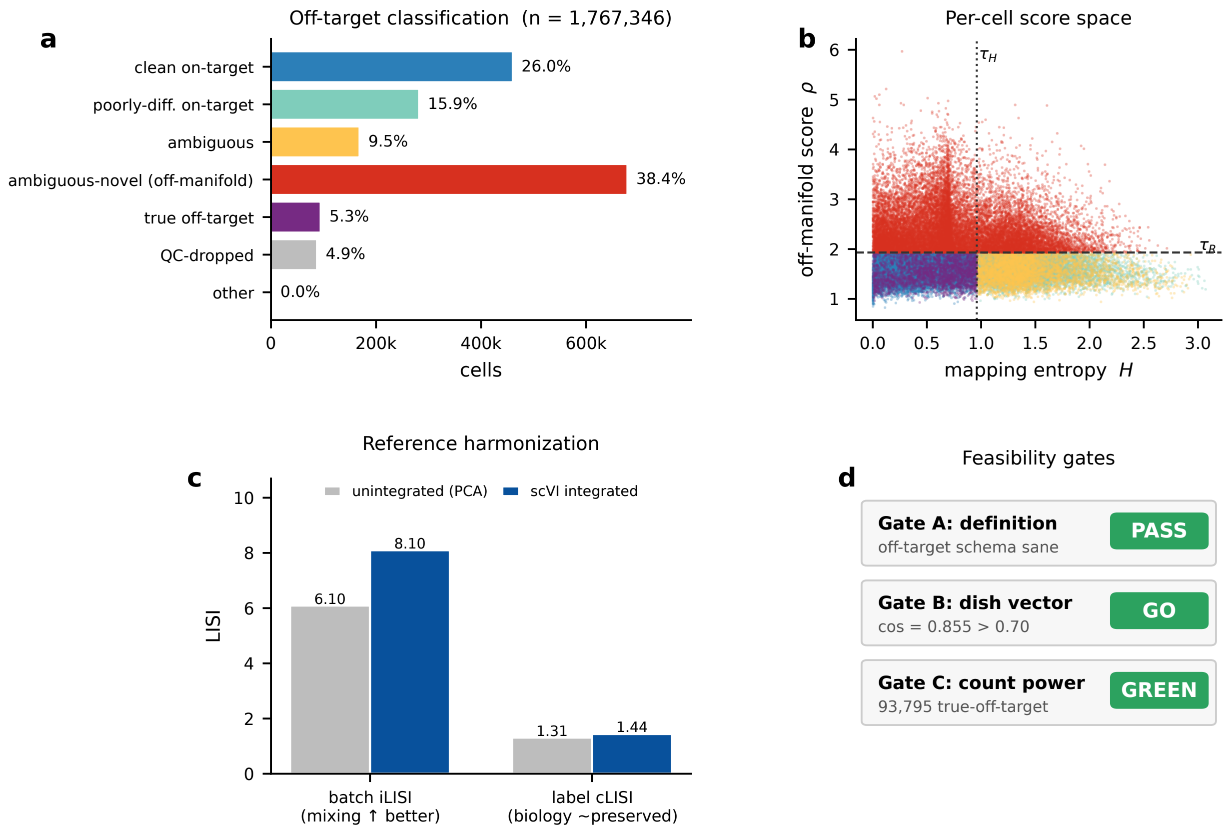

Figure 2: **Benchmark landscape.** (**a**) Query strata. (**b**) The two per-sample scores, with calibrated thresholds; the strata are regions of this plane. (**c**) Donor-level integration improves cross-batch mixing while preserving structure. (**d**) Sanity checks on the mapping. The *off-manifold* stratum (38.4%) is analyzed in Section 7; the *off-target* stratum (5.3%) supplies the multi-population geometry of Sections 5–6.

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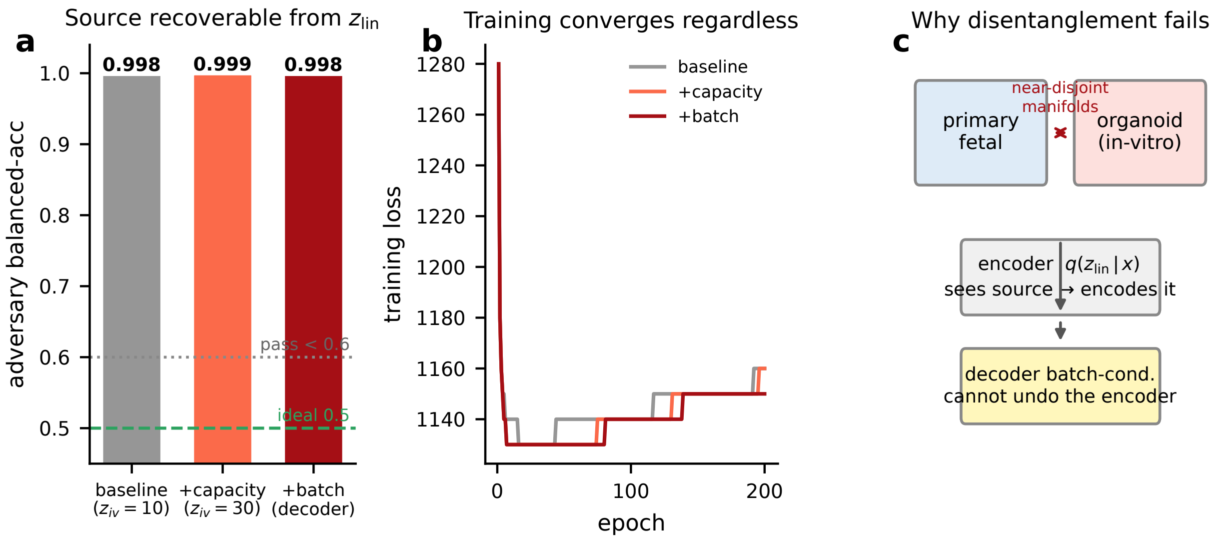

Figure 3: **Result 1: anchor-free invariance is not identified.** (**a**) A held-out adversary recovers the domain from the shared latent at ≥ 0.998 balanced accuracy, unchanged by tripling the salient capacity or by conditioning the decoder on domain (0.5 is invariance; < 0.6 would indicate success). (**b**) Training converges in every run, so the failure is structural rather than optimization-related. (**c**) Mechanism: with near-disjoint supports the encoder observes the domain directly, and decoder-side conditioning cannot remove what the encoder has already written.

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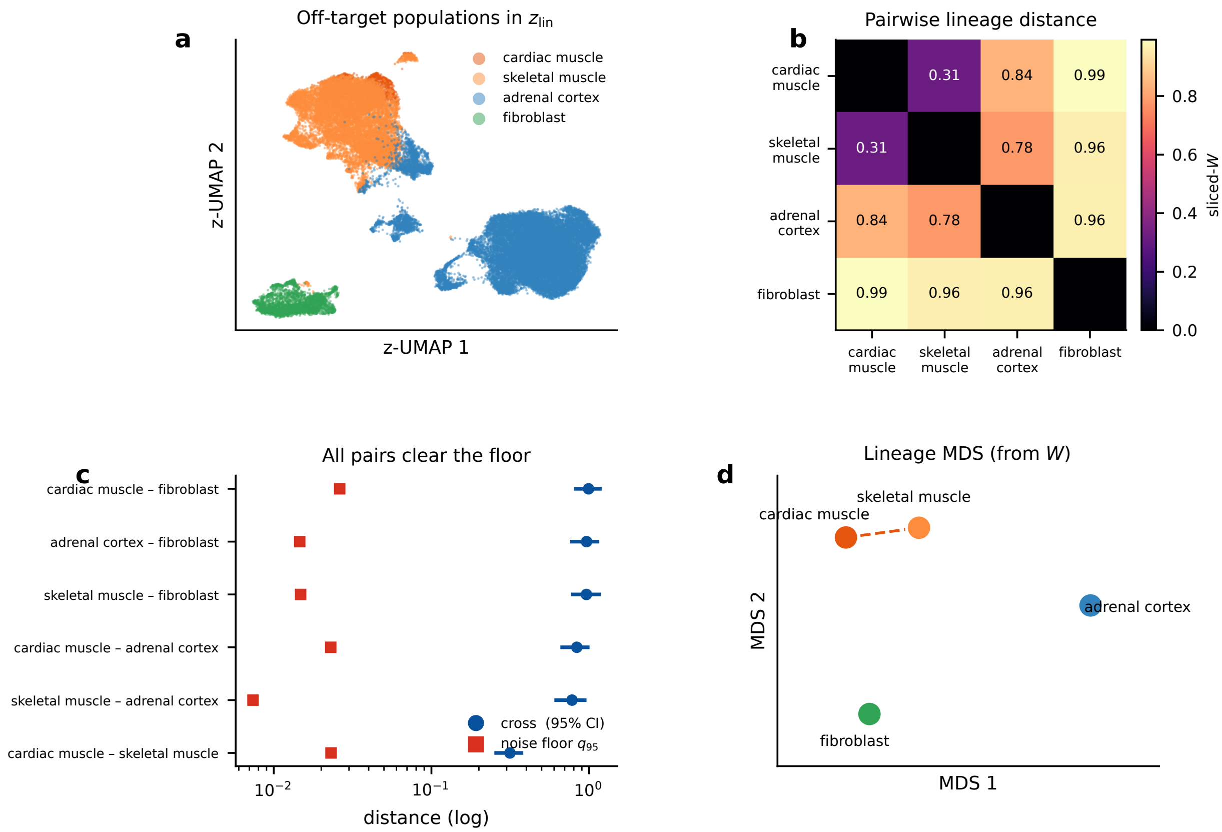

Figure 4: **Results 2 and 3: translation-invariant functionals, and the limits of the cancellation.** (**a**) The four well-powered populations in the shared latent. (**b**) Pairwise sliced-Wasserstein distances. (**c**) Every well-powered pair clears its matched-N noise floor by 14 to 105×. (**d**) Multidimensional scaling of the distance matrix. The *ordering* reproduces the reference geometry (distance-matrix $r = 0.90$), whereas the individual distances are confounded by population-dependent offsets (mean offset cosine 0.15; Section 6).

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A Benchmark construction details

Reference assembly, gene-space harmonization, quality control, the stabilizers required for the query projection at this scale, threshold calibration for the two per-sample scores, and the full stratum definitions are provided in the supplementary material, together with run logs, derived data, and code sufficient to reproduce every number reported above.

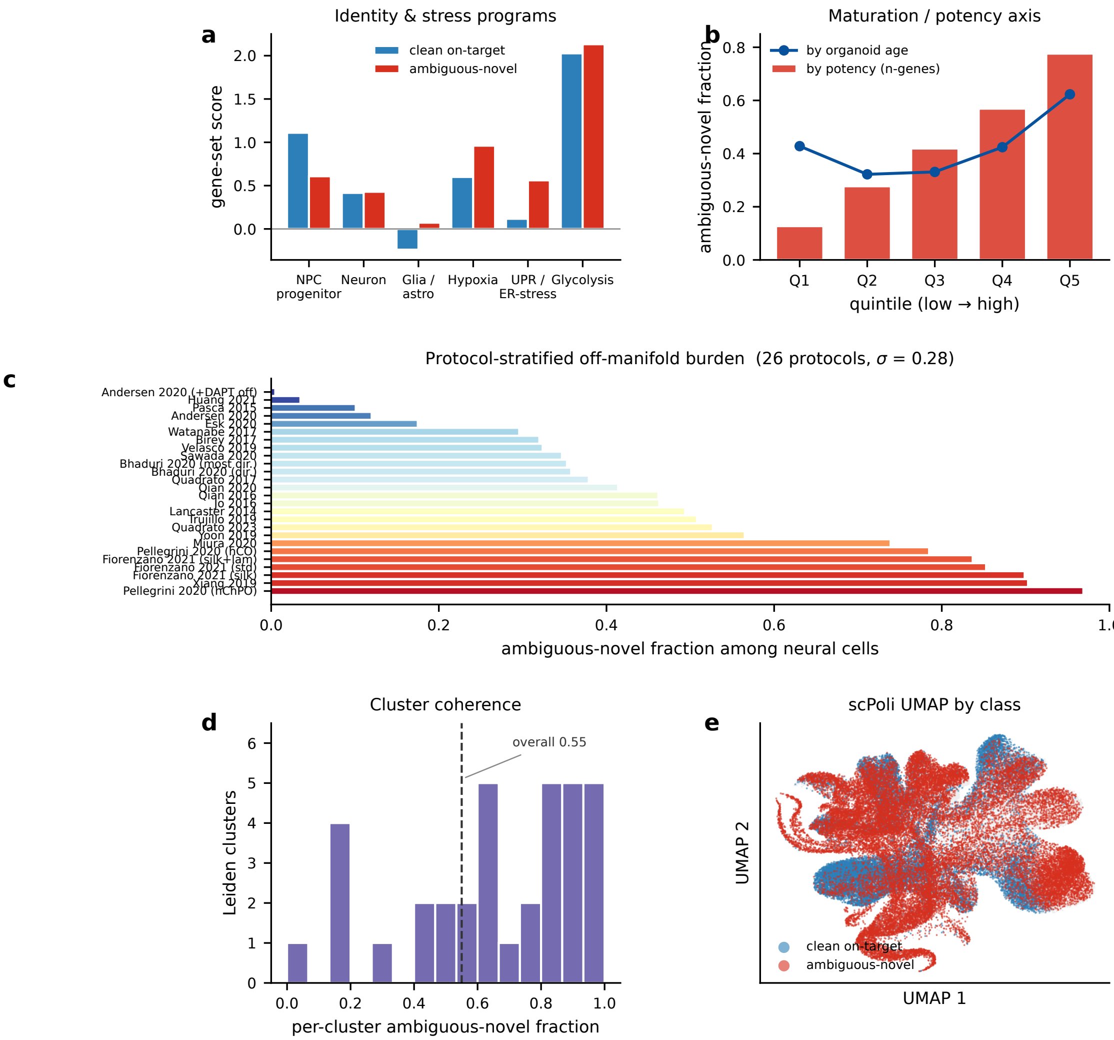

Figure 5: **The off-manifold stratum is structured, not noise.** (**a**) Reduced canonical progenitor identity with elevated stress programs. (**b**) Frequency rises monotonically with a per-sample complexity proxy. (**c**) Nearly 240-fold variation across 27 protocols (0.4% to 96.8%): protocols targeting states the reference covers score low, those targeting states it lacks score high, the signature of reference incompleteness. (**d**,**e**) The stratum concentrates in coherent clusters and occupies contiguous territory rather than scattering.

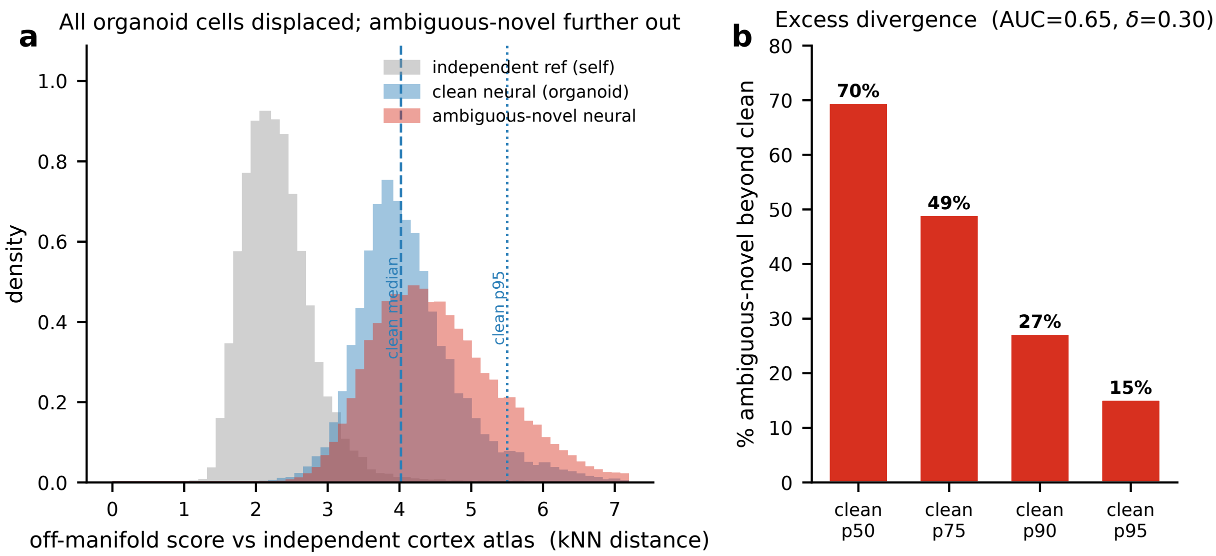

Figure 6: **Result 4: separating reference incompleteness from genuine divergence.** (**a**) The domain offset displaces *every* query sample from an independent reference, including the confidently-mapped control (medians 4.0 and 4.4 versus 2.2 within reference), so absolute "mapped-on" fractions are uninterpretable and are not reported. (**b**) Calibrating against the control removes the shared displacement: off-manifold samples are further from the independent reference than controls (AUC 0.65, Cliff's $\delta = 0.30$), confound-controlled evidence of genuine divergence over and above reference incompleteness.